

Chapter 36 — Pediatric Urogenital Imaging — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **ALARA** (as low as reasonably achievable): treat every child as radiation sensitive.
- **Diuretic renography (MAG3, mercaptoacetyl triglycine)** = gold standard, obstructive vs non-obstructive hydronephrosis.
- **Furosemide 1 mg/kg** only when dilated system **maximally filled**.
- **No paediatric T½ cut-off** defines obstruction → serial scans, curve shape, upright drainage.
- **Bilateral hydroureteronephrosis + thick bladder + poor emptying** → VCUG (voiding cystourethrogram) for PUV (posterior urethral valve).
- **Keyhole sign** (thick bladder + dilated posterior urethra) → PUV.
- **DMSA** (dimercaptosuccinic acid) lesion at **≥6 months** = fixed scar; APN (acute pyelonephritis) keeps contour, scar loses it.
- **No ultrasound for routine undescended testis** (Choosing Wisely); exception obese child.
- **Suspected torsion** → **explore**; never delay for imaging.
- **Blood at meatus** → **retrograde urethrogram before catheter**.

TECHNIQUE

- **Sedation risk > contrast risk** (NSF — nephrogenic systemic fibrosis; contrast nephropathy).
- **Fetal MR**: avoid in first trimester (gene-expression concern).
- **Ultrasound**: safe at all ages and all gestations.
- **Ionising harm greater in children**: radiosensitive tissue + lifetime cumulative dose.
- **Guideline bodies**: AAP, NICE, CPS, PSPN, EAU/ESPU; ACR Appropriateness Criteria.
- **MR hazards**: radiofrequency heating/hyperthermia (rare); gadolinium NSF.
- **Risk literature**: check estimated vs modelled vs measured risk.

ULTRASONOGRAPHY

- **Amniotic fluid**: fetal skin dialysate <8–10 wk; fetal urine from 10–12 wk.
- **Most anomalies detectable** by 20 wk; bladder visible by 10–12 wk.
- **Fetal bladder never fully empty**; residual always.
- **Normal cortex**: isoechoic/slightly hypoechoic to liver.
- **Prenatal ultrasound limited** by oligohydramnios, habitus, bowel gas, fetal position → MRI.
- **Low cord insertion/abdominal wall defect** → exstrophy/cloacal exstrophy.
- **Hydronephrosis underestimated** ≤2 days postnatal and in dehydration.
- **SFU (Society for Fetal Urology) grades**: 1 pelvis → 2 major calyces → 3 minor calyces → 4 + parenchymal thinning.
- **Always review images** — reporting systems vary.
- **Transition dilated** → **non-dilated** localises obstruction: UPJ (ureteropelvic junction), UVJ (ureterovesical junction) or urethra.
- **Megaureter/ureterocele**: obstructive or not → renography or MRU (MR urography).
- **Cysts don't communicate**; dilated calyces do.
- **MCDK** (multicystic dysplastic kidney): unilateral, random variable-size cysts.
- **ARPKD**: bilateral, enlarged homogeneously hyperechoic.
- **ADPKD**: bilateral, parenchyma replaced by cysts.
- **Multilocular cystic nephroma** = infancy; **cystic Wilms** = 2–4 yr.
- **Neonate + AKI (acute kidney injury) + hypertension + big kidney** → renal vein thrombosis.
- **Suprarenal neonatal mass** → adrenal haemorrhage; neonatal renal mass → mesoblastic nephroma.
- **Renal mass**: ultrasound + age/presentation → contrast CT.
- **Stones**: **ultrasound first**, not reflex CT (computed tomography)/KUB (kidney–ureter–bladder film).
- **Distal ureteric stones** easier to see in children than adults.
- **Stone** = echogenic + shadow; **fat** = echogenic, no shadow; twinkling on Doppler.
- **Hospitalised UTI**: ultrasound before discharge to exclude obstruction.
- **Normal ultrasound** ≠ no VUR; ≠ risk-stratified APN.
- **UTI failing to improve on antibiotics** → ultrasound.
- **CeVUS** (contrast-enhanced voiding urosonography): sulfur hexafluoride microspheres; AUC (area under curve) 0.97; false-negative 3%.
- **CeVUS uses**: VUR screening/follow-up; post-transplant; video-urodynamics.
- **Intrarenal reflux**: commonest lower pole; ↑ with VUR grade; linked to scarring.
- **Contrast ultrasound** = **contrast CT** sensitivity, blunt abdominal trauma.
- **Trauma follow-up**: ultrasound for perinephric haematoma/urinoma resolution.
- **Trauma work-up incomplete** without delayed CT views or lateral bladder/urethral projections.
- **Clot vs tumour**: supine/prone — clot moves.
- **Midline pelvic cyst**: hydrometrocolpos, ovarian cyst, bladder, urinary ascites, utericle.
- **Hydrometrocolpos**: imperforate hymen, vaginal atresia, transverse septum, urogenital sinus/cloaca.
- **Dilated utricle**: midline, posterior to male bladder.
- **Oligohydramnios + dual pelvic cysts below bladder** → cloaca (hemivaginas).
- **Viral cystitis**: upper respiratory prodrome + gross haematuria + frequency; thick wall mimics mass.
- **Exstrophy**: no bladder, low cord, short phallus.
- **Prune belly vs PUV**: empty scrotum, no thick bladder/keyhole.
- **Whirlpool sign**: sensitivity ~96% vs ~76% colour Doppler.
- **Appendage torsion**: avascular upper-pole nodule + surrounding hyperaemia.
- **No arterial flow + heterogeneous parenchyma** → non-viable testis.
- **Fountain sign** → acute idiopathic scrotal oedema.
- **Scrotal ultrasound overused** — history/examination diagnose most; don't overrule examination.
- **Commonest prepubertal testis tumour**: teratoma (solid/cystic/calcified).
- **Genital imaging, four indications**: acute scrotum, masses, cryptorchidism, ambiguous genitalia.

- **Yolk sac tumour:** homogeneous, vascular; **epidermoid:** onion rings, no flow.
- **CAH** (congenital adrenal hyperplasia): bilateral testicular adrenal rests.
- **DSD/non-palpable gonads + hypospadias:** ultrasound first for müllerian structures.

MAGNETIC RESONANCE IMAGING

- **Sedation:** usually needed <6 yr; protocol ~1 h (short ~30 min).
- **MRU furosemide:** 1 mg/kg, max 20 mg; gadolinium.
- **Fetal MRI** with third-level obstetric ultrasound after ~20 wk.
- **MRU:** no radiation; equivalence to renography unproven.
- **Rhabdomyosarcoma:** low T1, heterogeneous high T2; no subtype prediction.
- **MRI beats CT:** synchronous contralateral renal lesions.
- **Renal tumour staging:** MRI locoregional; CT (some) for distant metastasis.
- **Bladder MRI:** urine masks T2 lesions → T1 before gadolinium.
- **MRU function:** Patlak method; complex post-processing limits uptake.
- **No role:** acute scrotum, scrotal masses.
- **Undescended testis:** MRI marginally better than ultrasound at best.
- **DSD:** ultrasound + genitography suffice; MRI for very complex only.
- **Avoid sedation:** breath-hold/respiratory triggering (longer), child life specialists.

CONVENTIONAL RADIOGRAPHY

- **IVP** (intravenous pyelography): obsolete.
- **Retrograde pyelogram:** needs anaesthesia; intraoperative localisation.
- **KUB for constipation:** little evidence; persuades sceptical parents.

VOIDING CYSTOURETHROGRAPHY

- **After UTI:** VCUG once improving on antibiotics — no 4–6-week wait.
- **Early VCUG in febrile UTI** may overgrade VUR (endotoxin).
- **Hydronephrosis:** VCUG only if dilated ureter(s).
- **Catheter off midline:** utricle/posterior urethra (male), ectopic ureter (female).
- **Bladder capacity:** (age + 2) × 30 mL, max 400 mL.
- **VCUG masses:** urachal/bladder diverticula, dilated utricle, fibroepithelial polyp.
- **Upper-tract malformation** benefiting from VCUG: UVJ anomalies only.
- **Occult VUR:** contralateral new VUR 7–33%; VCUG finds VUR in ~40% of DMSA-proven APN.
- **PIC** (positional instillation of contrast): cystoscopic, 1 m; negative VCUG + recurrent febrile UTI.
- **Any UTI in circumcised boy** may merit evaluation (risk very low).
- **Recurrent epididymitis in boys** → VCUG (ectopic ureter, utricle).
- **Urogenitogram** → Shopfner; low urethrovaginal confluence = type III.
- **Pie-in-the-sky** bladder + extraperitoneal contrast → pelvic-fracture bladder injury.
- **VCUG scout film:** spinal abnormalities.

COMPUTED TOMOGRAPHY

- **Chosen for speed, not accuracy:** blunt trauma, polytrauma.

- **Arterial phase only** misses collecting system/ureter → delayed images.
- **CT cystography:** classifies intra- vs extraperitoneal rupture.
- **Paediatric renal tumours:** contrast CT routine.
- **CT/MR for UTI:** abscess, fistula, postoperative, non-ionising inconclusive.
- **Stone CT:** better burden estimate, rarely changes management.
- **CT scrotum:** no role except testicular tumour staging.

DMSA RENAL CORTICAL SCINTIGRAPHY

- **Tubular uptake,** flow-dependent; APN = photopenia.
- **DMSA timing:** days for APN; ≥6 months for scar.
- **DMSA for APN:** sensitivity 87%, specificity 100% (piglet model).
- **Acute lesions persist** in 36–52%; resolve over ~5 months.
- **DMSA beats ultrasound** for APN and scarring.
- **Top-down:** DMSA first, VCUG only if abnormal; **bottom-up:** VCUG → overtreats low-grade VUR.

MAG-3 DIURESIS RENOGRAPHY

- **Four elements:** hydration, ROI (region of interest)/background, bladder catheter, diuretic timing.
- **T½ from maximal filling,** not tracer injection.
- **Progression predictors:** biphasic curve; poor upright drainage.
- **Dietl's crisis:** cortical tracer retention up to 48 h.
- **Catheter:** removes bladder-filling effect, VUR confusion, gonadal dose.
- **Wrong ROI** → spurious function/drainage; ROI can localise UPJ vs UVJ.
- **Renography vs ultrasound:** more reproducible, quantifiable; SFU 3 can hide high-grade UPJ obstruction.

RADIONUCLIDE CYSTOGRAPHY AND TESTICULAR SCANNING

- **Direct:** more sensitive, less radiation than VCUG; poor anatomy; grading hard to match.
- **Indirect:** no catheter, insensitive → not routine.
- **Testicular scan** (^{99m}Tc-pertechnetate): cold = torsion, hot = inflammation; ≤100% accurate; only if no surgical delay.

CLASSIC TRAPS

- Trap: neonatal dark pyramids = hydronephrosis — Actually: normal newborn kidney.
- Trap: day-1 scan excludes hydronephrosis — Actually: underestimates ≤2 days.
- Trap: dilated ureter = obstruction — Actually: may be VUR.
- Trap: normal ultrasound excludes VUR — Actually: poor predictor.
- Trap: T½ from tracer injection — Actually: from maximal filling.
- Trap: dehydrated renogram shows obstruction — Actually: poor diuresis mimics it.
- Trap: one normal renogram excludes obstruction — Actually: may progress; repeat.
- Trap: free retrograde flow = patent — Actually: flap may block antegrade.
- Trap: hyperaemic epididymis = epididymitis — Actually: also post-detorsion.
- Trap: arterial-phase CT clears collecting system — Actually: needs delayed phase.